

CSI-260: Final Exam Instructions

Question 1: (5 pts.)

Open the file Q1_list_mgr.py in PyCharm. In this file, create a list comprehension named `sq_rts` that generates the square root of a whole number between 1 and 100. Print only those square roots that are perfect square roots.

Question 2: (35 pts.)

Open the file Q2_garbage.py in PyCharm. There is a function in this file named `q2Output()`. In this function, write code that will separate out the data types so that when you are done, the list `lst_stuff` contains only integer data, the list `str_stuff` contains only string values, the list `fl_stuff` contains only floating-point values and the list `bool_stuff` contains only Boolean values.

Create an empty array named `int_array` and populate this array with a 4 by 4 array of integers. Next calculate the following metrics of this array:

- a) average of the integers
- b) median value
- c) variance
- d) standard deviation

Write code in the function that will output the results/contents of:

1. `lst_stuff`
2. `str_stuff`
3. `fl_stuff`
4. `bool_stuff`
5. `int_array`
6. all metrics of the array `int_array`

Question 3: (25 pts.)

Open the Excel file named `"Seminars.xlsx"` and examine its contents. Create a Python file named `"seminars.py"`. In your Python file, create a function named `mgr_data()`: and in this function have it do the following:

1. add a new worksheet to the `Seminars.xlsx` file named `"calculations"`

In this new worksheet:

2. average the costs of running the “Account” seminars
3. total the “Seminar Prince per Person”

The worksheet should look like the image below:

Average	\$961
Total Seminar Price per Person	\$60,605

Question 4: (25pts.)

In PyCharm, create a new file and save it as “CarClasses”. In this file, you will create a class named Cars. This class has several characteristics, the first three are “make”, “model” and “year”. The make and model attributes take only string data and the year takes only integer data. The next attributes for this class include “type”, which is a string data type and “mpg” which is an integer data type. The final attributes of the Car class are “cylinder”, “turbo” and “hybrid”. The “cylinder” attribute is an integer data type while the remaining attributes are both Boolean data types.

The next step is to create a file named “Car.py” and save it in the same location as the CarClasses.py file. In the Car.py file, write code that will assign the following data to each car:

Make	Model	year	type	cylinder	turbo	hybrid
Ford	EcoSport	2022	unleaded	4	no	no
Toyota	Corolla	2024	unleaded	6	no	yes

When you are done, print this data to the output console. You can use a “print()” method or incorporate a “return” in your code, either is fine. It does not have to look like the data presented above.

Question 5: (10 pts.)

Using your knowledge of *args and **kwargs, create a function that will take a variable input of factors named mathFunction(). In this function, set it up so that it will any number of integers and print the addition and multiplication of the integers fed to the function.

As a part of this question, you must feed the dictionary below to the function and have it return the addition and multiplication of the values.

```
math_values1 = {'one': 10 'two': 22, 'three': 7, 'four': 45, 'five': 20}
```

```
math_values2 = [2, 4, 8, 16, 32, 64, 128]
```

```
math_values3 = [ 0, 1, 1, 2, 3, 5, 8, 13]
```

You are free to use f-strings, the `print()` method or `return` to manage your output.

Please save all work and upload the exam solutions to Canvas.